

FAKE NEWS DETECTION

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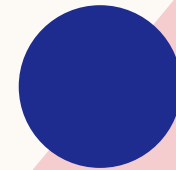
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BUSINESS UNDERSTANDING

The ability to automatically distinguish fake from real news is essential for:

- Social media platforms
- Fact-checking organizations.
- News aggregators
- The general public.



PROBLEM STATEMENT

- The proliferation of fake news on digital platforms poses a significant risk to public trust and decision-making.
- Traditional methods of identifying fake news are slow, and resource-intensive.

PROPOSED SOLUTION

- Develop an automated binary classification system using transformer-based Natural Language Processing (NLP) models to detect and flag fake news articles.

Potential Impact: Avail a scalable solution to help identify harmful misinformation, improve information credibility and support fight against digital manipulation.



OBJECTIVES

1. Perform Exploratory Data Analysis.
2. Build a baseline Logistic Regression model to predict an article's class (fake or real).
3. Build an XGBoost, a LSTM neural network and finetune the RoBERTa-base transformer.
4. Explain the best-fit model's predictions at feature level.
5. Deploy selected model.

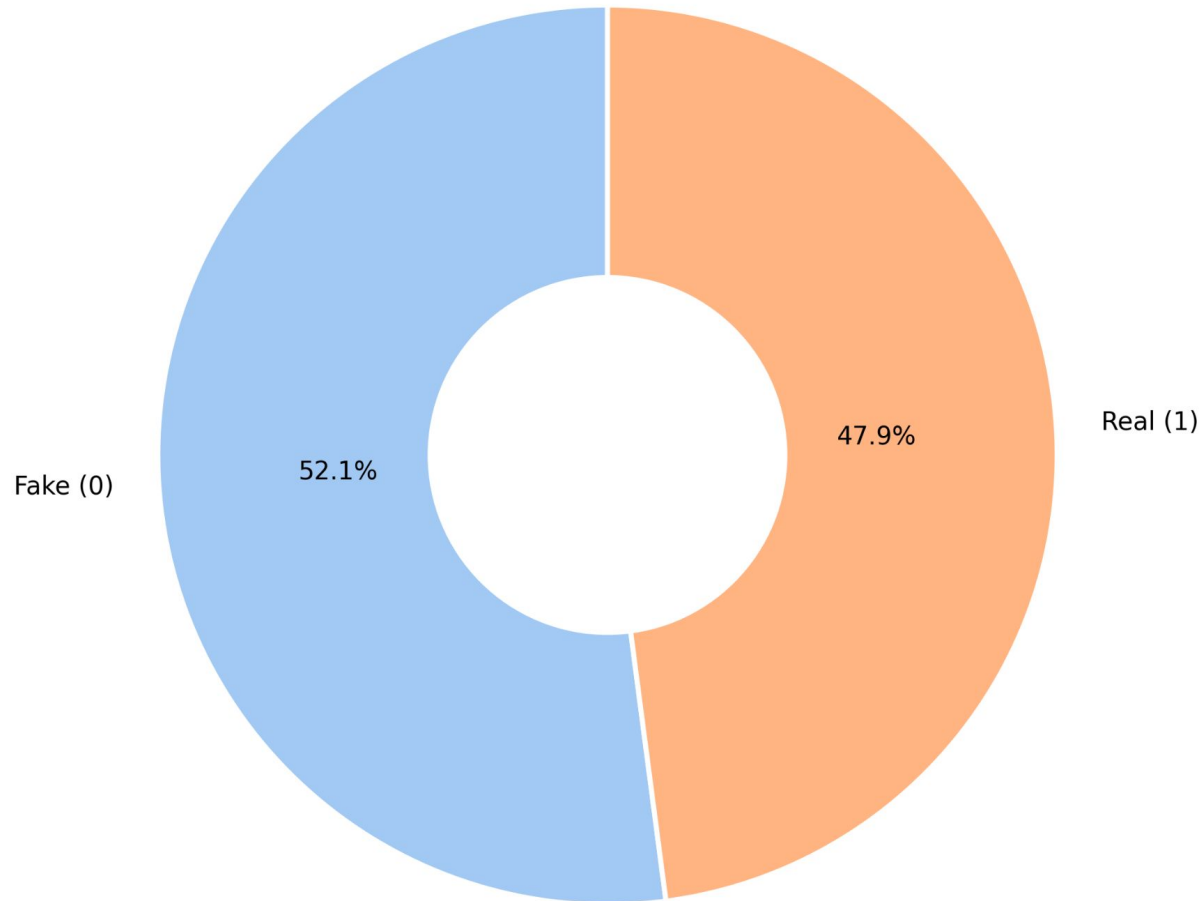
DATASET

Datasets used;

- ❖ gossipcop_fake.csv
 - ❖ gossipcop_real.csv
 - ❖ politifact_fake.csv
 - ❖ politifact_real.csv
 - ❖ kaggle/Fake News Detection /fake.csv
-
- The body text for news articles across the 5 datasets was scrapped based on respective `news\_url`.
 - The enriched datasets were concatenated to yield a working dataset with 22,465 entities.

FINDINGS: EDA

Distribution of Class Feature



The data is highly imbalanced:

- Fake articles = 47.9%
- Real articles = 52.1%

Strategies to address imbalance:

- Oversample minority class (SMOTE).
- Undersample majority class (RUS).

To determine the most impactful strategy; each ML model was trained using:

- 1) The imbalanced dataset.
- 2) SMOTE-balanced data.
- 3) RUS-balanced data.

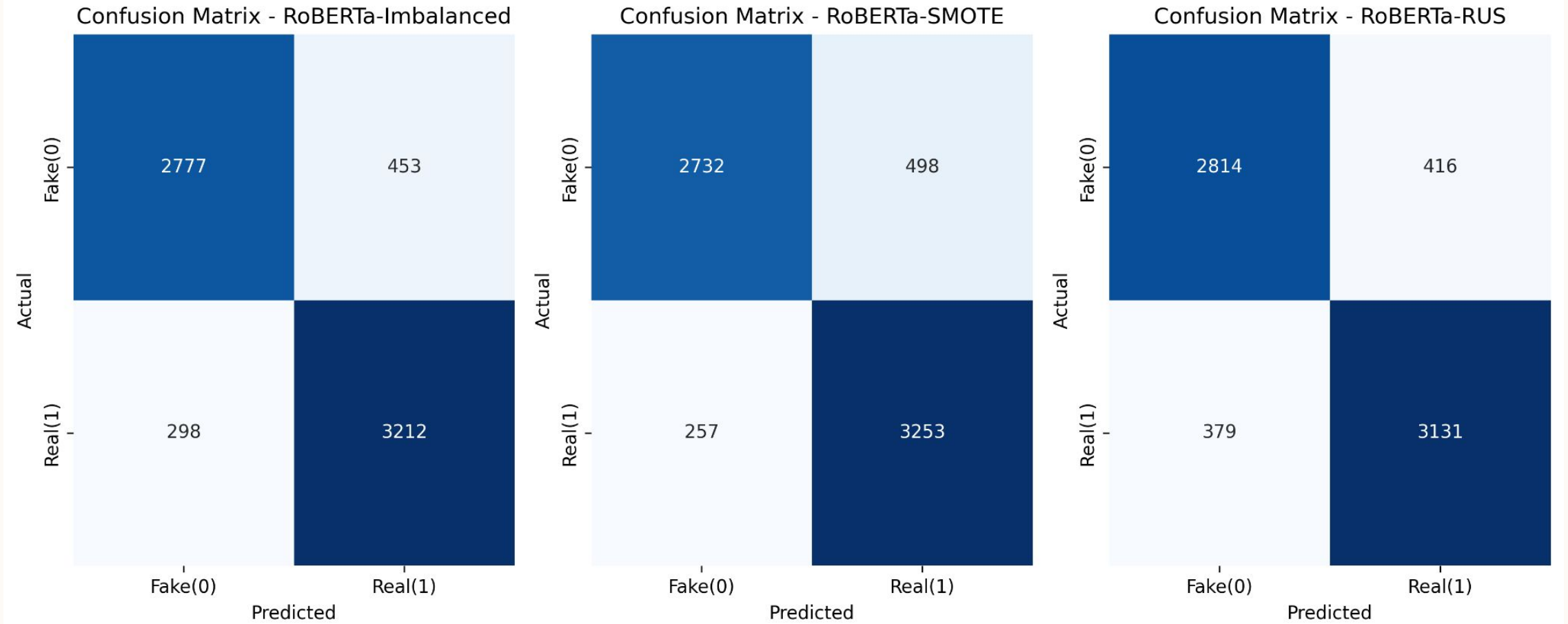
FINDINGS: MODELLING

Model	Best Configuration	Macro-averaged Recall	Recall (Fake)	Recall (Real)	Precision (Fake)	Precision (Real)	Accuracy	AUC
Logistic Regression	RUS-balanced data	0.8614	0.8214	0.9014	0.8846	0.8458	86.31%	0.930
XGBoost Ensemble model	RUS-balanced data	0.8741	0.8427	0.9054	0.8913	0.8622	87.54%	0.938
LSTM Neural Network	Imbalanced data	0.8277	0.7130	0.9425	0.9194	0.7811	83.25%	0.939
Fine-tuned RoBERTa-base	RUS-balanced data	0.8816	0.8712	0.8920	0.8813	0.8827	88.2%	0.951

The fine-tuned RoBERTa-base transformer is the best-fit, and most generalizable model because it achieves:

- ❖ **Highest Recall (Fake):** Catches most fake articles.
- ❖ **Macro-averaged Recall:** Strikes an effective balance in catching both real and fake articles.
- ❖ **Highest Precision (Real):** Highest certainty for class 1 predictions when an article is actually real.
- ❖ **Highest Accuracy:** Most accurate across either prediction.

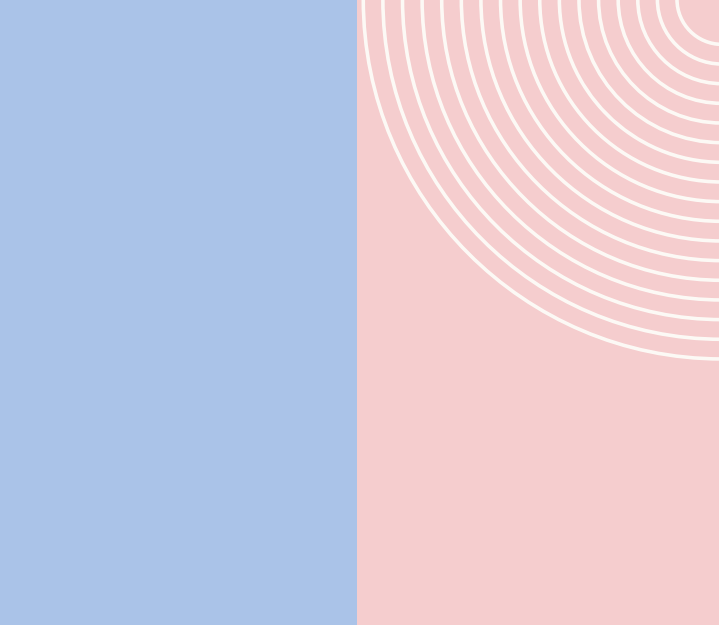
FINDINGS: FINE-TUNED RoBERTa-base



→ The RoBERTa-base transformer fine-tuned with Undersampled-balanced data performs better on accurate predictions for “Fake News Articles”.

CONCLUSIONS

- ❖ **Fake News Detection is possible:** We created a system that can effectively identify fake news articles.
- ❖ **Advanced Technology for better accuracy:** The system uses RoBERTa, which understands language to spot fake news accurately.
- ❖ **Balanced Approach:** The model balances fake and real news to improve detection without losing accuracy.
- ❖ **Continuous Improvement:** The system adapts over time to avoid overfitting and stay accurate.
- ❖ **Real-World Use:** The system can be used in apps or browser extensions for real-time fake news detection.
- ❖ **Expansion Potential:** The technology can be expanded to other languages and platforms.



NEXT STEPS

- ❖ Deploy the containerized fine-tuned RoBERTa model via AWS SageMaker for low-latency inference.
- ❖ Develop browser extensions and social media plugins using the model's API for real-time credibility scoring.
- ❖ Incorporate automation frameworks to flag low-confidence predictions for human review.





THANK YOU